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Development, Molecular Biology, and Physiology of the Prostate

Brian W. Simons, and Ashley E. Ross

Keywords

androgen receptor; androgen; cell biology; differentiation; hormone; prostate development; prostate-specific antigen; prostate-specific membrane antigen; seminal vesicle; steroid

Questions

1. Which of the following is NOT considered a sex accessory tissue?
    a. Prostate gland
    b. Seminal vesicles
    c. Tunica albuginea
    d. Ampullae
    e. Bulbourethral gland
2. Which fetal hormone stimulates the development of the wolffian ducts?
    a. Estradiol
    b. Dihydrotestosterone (DHT)
    c. Estrone
    d. Testosterone
    e. Inhibin
3. Which one of the following statements about fetal development of the lower urogenital tract is FALSE?

a. The urogenital sinus derives from the cloaca.
b. The cloaca gets its name (L. “sewer”) because it receives input from the gastrointestinal and urinary tracts.
c. The seminal vesicles derive from the posterior portion of the wolffian ducts, whereas the prostate derives from the anterior portion.
d. Sox9 is an early molecular marker of prostate development.
e. The urogenital sinus is a primordial structure that contributes to bladder *and* prostate development.

4. Which fetal hormone is most important in stimulating the growth of the prostate during development?
   a. Estradiol
   b. DHT
   c. Estrone
   d. Testosterone
   e. Inhibin
5. Which one of the following statements is TRUE regarding the role of androgens in prostate development?
   a. Males will develop prostates in the presence of sufficiently high levels of androgens, but females will not.
   b. Females will develop prostates in the presence of sufficiently high levels of androgens, but males will not.
   c. Both males and females will develop prostates in the presence of sufficiently high levels of androgens.
   d. Prostate tissue rudiments with a normal androgen receptor in the epithelium but mutant androgen receptor in the mesenchyme will develop into normal prostates in the presence of sufficiently high levels of androgen.
   e. Prostate tissue rudiments with androgen receptor overexpression in the epithelium but mutant androgen receptor in the mesenchyme will develop into normal prostates in the absence of androgen.
6. Which α_1-adrenergic receptor subtype is linked to smooth muscle contraction in the prostate?
   a. α_D

b. α_{1A}
c. α_{1B}
d. α_2
e. α_{2B}

7. Which of the following is TRUE regarding testosterone?
   a. Testosterone is synthesized by the Sertoli cells of the testes.
   b. Testosterone is synthesized by the Leydig cells of the testes.
   c. Testosterone is a direct precursor of pregnenolone.
   d. 5α-Reductase is an enzyme that converts DHT into testosterone.
   e. Aromatase converts estrogens into testosterone.
8. Dehydroepiandrosterone (DHEA) has been suggested as a major source of testosterone within the plasma. What percentage of total testosterone has been determined to be derived from DHEA?
   a. 1%
   b. 2%
   c. 5%
   d. 15%
   e. 20%
9. To what is the majority of testosterone in the plasma bound?
   a. Insulin
   b. Cholesterol
   c. Prostaglandins
   d. TP53
   e. Sex hormone-binding globulin (SHBG)
10. Which 5α-reductase isoform predominates in the prostate gland?
   a. Type 1
   b. Type 2
   c. Type 3
   d. Type 4
   e. Type 5
11. What is the source of fructose in seminal plasma?
   a. Prostate gland
   b. Bulbourethral gland
   c. Vas deferens

d. Seminal vesicles
e. Basal cells

12. Compartmentalization of the androgen receptor from the cytosol to the nucleus is dependent on:
    a. dimerization.
    b. adenosine triphosphate (ATP).
    c. RAN-mediated transport.
    d. nuclear localization and nuclear export signals.
    e. all of the above.
13. Which feature of the androgen receptor polymorphisms is thought to impact on the overall activity of androgen target gene induction?
    a. Zinc finger motifs
    b. Poly CAG repeats
    c. Nuclear export signals
    d. Ligand binding domain
14. Which of the following is TRUE with regard to the effects of estrogens in prostate development?
    a. Estrogens are not required for the development of a prostate because ER-α and ER-β knockout mice have phenotypically normal prostates.
    b. ER-α regulates ductal formation.
    c. ER-β is required for prostates to allow sperm to mature to an active form.
    d. b and c
15. When the androgen receptor binds to an androgen response element, which of the following is (are) true?
    a. The dimerization of the receptor is always head to head, regardless of whether the sequence is a direct repeat or an inverted repeat.
    b. The dimerization occurs head to head for an inverted repeat and head to tail for a direct repeat.
    c. The dimerization of the androgen receptor occurs on the DNA template in a process that requires ATP and heat shock proteins.
    d. The androgen receptor can bind to a target androgen response element in either a head-to-head or a head-to-tail orientation,

depending on the orientation of the structural gene.

Answers

1. **c. Tunica albuginea.** Sex accessory tissues include the prostate gland, seminal vesicles, ampullae, and bulbourethral glands. They are believed to play a major, but unknown, role in the reproductive process.
2. **d. Testosterone. The wolffian ducts develop into the seminal vesicles, epididymis, vas deferens, ampulla, and ejaculatory duct; the developmental growth of this group of glands is stimulated by fetal testosterone and not DHT .**
3. **c. The seminal vesicles derive from the posterior portion of the wolffian ducts, whereas the prostate derives from the anterior portion. The prostate develops from the urogenital sinus, not the wolffian duct.**
4. **b. DHT.** The prostate first appears and starts its development from the urogenital sinus during the third month of fetal growth, and development is directed primarily by DHT, not testosterone.
5. **c. Both males and females will develop prostates in the presence of sufficiently high levels of androgens. If androgen (testosterone or DHT) levels are sufficiently high at the right time in fetal development, prostate development will proceed in the urogenital sinus regardless of whether the embryo is male or female. For prostate development to proceed, androgen receptor is required to be functional in the mesenchyme.**
6. **b.** α_{1A}. **Research work has demonstrated three subtypes of the α_1-adrenergic receptor (α_{1A}, α_{1B}, and α_{1D}), of which the α_{1A} receptor appears to be linked to contraction .**
7. **b. Testosterone is synthesized by the Leydig cells of the testes.** Foremost among the hormones and growth factors that stimulate the prostate is the prohormone testosterone, which must be converted within the prostate into the active androgen DHT. **Testosterone is synthesized in the Leydig cells of the testes from pregnenolone by a series of reversible reactions; however, once**

testosterone is reduced by 5α-reductase into DHT or to estrogens by aromatase, the process is irreversible. In other words, whereas testosterone can be converted into DHT and into estrogens, estrogens and DHT cannot be converted into testosterone.

8. **a. 1%.** Less than 1% of the total testosterone in the plasma is derived from DHEA.
9. **e. Sex hormone-binding globulin (SHBG).** The majority of testosterone bound to plasma protein is associated with SHBG.
10. **b. Type 2. The type 2 isoform is mutated in 5α-reductase deficiency and is the dominant isoform present in the prostate gland.**
11. **d. Seminal vesicles. The source of fructose in human seminal plasma is the seminal vesicles.** Patients with congenital absence of the seminal vesicles also have an associated absence of fructose in their ejaculates.
12. **e. All of the above. After binding ligand, the androgen receptor dissociates from chaperonins, dimerizes, and then is transported to the nucleus via a nuclear localization motif, which activates Ran-dependent active transport, a process that is ATP dependent.**
13. **b. Poly CAG repeats. The shorter the length, the more actively the androgen receptor is thought to function.**
14. **a. Estrogens are not required for the development of a prostate because ER-α and ER-β knockout mice have phenotypically normal prostates.**
15. **a. The dimerization of the receptor is always head to head, regardless of whether the sequence is a direct repeat or an inverted repeat.** Although most models predicted that inverted repeat and direct repeat androgen response elements would have dimers that bind with opposite polarity, this prediction was not observed in x-ray crystallography studies.

Chapter Review

1. The seminal vesicles are extremely resistant to disease.
2. There are two major cellular components in the prostate: epithelial and stromal.
3. The stromal component consists of connective tissue, smooth muscle cells, and fibroblasts.
4. The epithelial component consists of basal cells, intermediate cells, and neuroendocrine cells.
5. The prostate contains a rich plexus of autonomic nerves.
6. Because of the diurnal variation of serum testosterone, to avoid inconsistency it should be measured in the morning.
7. The role of circulating adrenal androgens on prostate growth is minor.
8. Because the plasma half-life of testosterone is 10 to 20 minutes, patients who undergo bilateral orchiectomy are functionally castrate within 1 to 2 hours after surgery.
9. The androgen receptor is transported to the nucleus and then back to the cytoplasm; nuclear matrix is the major target for androgen and estrogen receptor binding in the nucleus.
10. The longer the CAG repeat in the androgen receptor, the lower its activity in activating target genes.
11. The source of prostaglandins, fructose, and semenogelin is the seminal vesicle.
12. The seminal vesicle contributes the most volume to the seminal fluid.
13. The source of citrate, zinc, spermine, and choline is the prostate.
14. Prostate-specific antigen is a serine protease and degrades semenogelin. Semenogelin gives rise to the coagulation of semen.
15. Complexed PSA is irreversibly bound to α1-antichymotrypsin and α2-macroglobulin.
16. Increases in human kallikrein 2, pro-PSA, and bound PSA are associated with prostate cancer.
17. Few drugs reach concentrations in the prostatic secretions that approach or surpass their concentrations in the serum. The exceptions include erythromycin, oleandomycin, sulfonamides, tetracycline, clindamycin, trimethoprim, and the fluoroquinolones.
18. Prostatic fluid is more acidic than is serum.

19. Acid phosphatase produced in the prostate may be elevated in prostate cancer; it is also produced in the bone and may be elevated in diseases that affect the bone such as Paget disease, osteoporosis, and bone metastases.
20. The wolffian ducts develop into the seminal vesicles, epididymis, vas deferens, ampulla, and ejaculatory duct; the developmental growth of this group of glands is stimulated by fetal testosterone and not DHT.
21. Testosterone is reduced by 5α-reductase into DHT or to estrogens by aromatase; the process is irreversible.

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Benign Prostatic Hyperplasia

Etiology, Pathophysiology, Epidemiology, and Natural History

Douglas W. Strand

Keywords

benign prostatic hyperplasia; epidemiology; lower urinary tract symptoms; natural history; pathophysiology; progression; prostate surgery; urinary retention

Questions

1. Which statement is correct regarding the role of androgenic hormones in the etiology of benign prostatic hyperplasia (BPH)?
    a. Testosterone and dihydrotestosterone (DHT) are the sole causes of the hyperplasia taking place in the prostate after the age of 40 years.
    b. The total amount of androgen receptors in the prostate decreases with aging, leading to a lesser response to androgenic stimuli.
    c. DHT is considered the more potent of the androgenic steroid hormones by a factor of approximately 10:1.
    d. Of the two 5α-reductase isoforms, type 1 is most commonly found in the prostate.

e. Only testosterone produced in the testis and not in the adrenal gland enters into the prostate gland.

2. Regarding genetic and familial factors in the etiology of BPH, which statement is TRUE?
    a. There is no evidence that BPH is a familial disease.
    b. Any man who has undergone transurethral resection of the prostate (TURP) should alert his sons that their chances of requiring TURP is three times greater than age-matched controls.
    c. Cases of familial BPH tend to occur in men with smaller prostates than the sporadic cases of BPH.
    d. Approximately 50% of cases of BPH in men who undergo surgery when younger than the age of 60 years are estimated to be inheritable.
    e. The most likely inheritance pattern is autosomal recessive.
3. The prevalence of a disease is defined as the number of:
    a. diseased people per 100,000 population per year.
    b. existing cases per 100,000 population at a distinct target date.
    c. deaths per 100,000 population per year.
    d. deaths per number of diseased.
    e. cumulative cases of a disease over a specified time period.
4. Concerning the autopsy prevalence of BPH or stromoglandular hyperplasia of the prostate, which statement is correct?
    a. No adequate studies have been done to date.
    b. It is commonly found in men of all ages.
    c. It is very uncommon in men younger than 30 years.
    d. It is found in 100% of men beginning at the age of 40 years.
    e. International comparisons are impossible because of a lack of its definition.
5. Which of the following statements regarding the International Prostate Symptom Score (IPSS) is TRUE?
    a. Moderate symptom severity is defined as a score from 12 to 22 points.
    b. The IPSS score addresses voiding and storage symptoms as well as questions regarding incontinence.

c. Quantitative symptom scores in BPH are not as important as are objective measures such as a flow rate recording.
d. The IPSS score has been translated and validated in many languages.
e. Physicians and nurses may fill out the IPSS score for their patients after consultation.

6. Which statement is TRUE regarding prostate volume?
a. International studies show significant similarity in prostate volume in white, age-stratified men.
b. Prostate volume assessment by digital rectal examination (DRE) is reproducible across examiners.
c. Although there is a steady increase in total prostate volume with age, the transition zone volume increases only marginally.
d. Magnetic resonance imaging (MRI) measurements are, in general, smaller compared with transrectal ultrasound measurements.
e. DRE estimation of prostate volume is fairly accurate when done by an experienced urologist.

7. Concerning liver disease and BPH, which of the following statements is TRUE?
a. Ethanol consumption increases circulating levels of estrogens.
b. The risk of having surgery for BPH is increased in heavy drinkers.
c. The intake of ethanol can decrease serum testosterone levels by a variety of mechanisms.
d. Most autopsy studies find a higher prevalence of BPH in men with liver cirrhosis.
e. In men with liver disease, histologic specimens of the prostate show a similar influence of estrogen such as seen in hormonally treated prostate cancer.

8. How do medications influence symptoms and flow rate?
a. There is no documented influence of any medication on symptoms or flow rate.
b. Antihistamines and bronchodilators significantly decrease urinary flow rates.

c. Calcium channel blockers and β-adrenergic blockers reduce urinary flow rates significantly.
d. Antidepressants, antihistamines, and bronchodilators increase the symptom score by several points.
e. Anticholinergic agents decrease the peak urinary flow rate markedly.

9. Concerning correlations between baseline parameters, which statement is TRUE?
   a. A clinically useful correlation exists between prostate volume and serum prostate-specific antigen (PSA) level.
   b. Many studies have shown a significant correlation between the transition zone volume and symptom severity.
   c. Correlation of symptoms, bother, interference, and quality of life are poor.
   d. Urinary flow rate and prostate volume correlate highly with serum PSA level.
   e. Serum PSA level shows a strong correlation with symptom frequency and bother.
10. Which statement is correct regarding the study of the natural history of BPH?
   a. Placebo groups from treatment trials are useful because they do not have treatment biases.
   b. A longitudinal population-based study has the fewest biases and is the most useful type of study.
   c. Control groups from intervention or medical therapy trials reflect the natural history of the disease in unselected community-dwelling men.
   d. Placebo groups have fewer selection biases compared with population-based studies.
   e. No such studies have been conducted.
11. Regarding the magnitude of the placebo response and its perception, which of the following statements is TRUE?
   a. Placebo response is not dependent on the baseline severity score.
   b. Most patients report subjective improvement when the drop from baseline is 30%.

c. The higher the baseline score, the more of a drop is required for patients to subjectively feel improved.
d. Perception of improvement is independent of baseline score.
e. There are convincing data to demonstrate that the final score after treatment is more important than the baseline score or the drop from baseline.

12. Descriptive studies of the incidence rates of acute urinary retention (AUR) have demonstrated that:
    a. depending on the population studied, incidence rates less than 5 to more than 130 cases/1000 man-years have been reported.
    b. the incidence rates reported do not differ significantly between various studies and populations.
    c. AUR has been poorly defined, and therefore no incidence rate can be calculated.
    d. incidence rates of approximately 10/1000 man-years have been reported in all watchful waiting studies.
    e. incidence rates of AUR have not been reported in the urologic literature, only prevalence rates.
13. What is the most significant finding regarding analytical epidemiology of AUR?
    a. Serum PSA level is a more powerful predictor of AUR than is age.
    b. Serum PSA level and prostate volume have limited ability to predict episodes of AUR.
    c. Urinary flow rates in placebo control groups are strong predictors of AUR episodes.
    d. Age has been found to be the most significant risk factor for AUR in population-based studies.
    e. There is virtually no relationship between symptom frequency and bother and AUR episodes.
14. Which statement regarding surgery for BPH is TRUE?
    a. The incidence rates of surgery are similar across wide geographic regions and ethnic backgrounds.
    b. AUR is a harder and more objective end point compared to surgery.
    c. Surgery is a less common end point compared with AUR.

d. Most patients with BPH eventually require surgery for their condition.
e. Surgery rates for BPH have remained stable since approximately 1990.

Answers

1. **c. DHT is considered the more potent of the androgenic steroid hormones by a factor of approximately 10:1.** The androgenic steroid hormones testosterone and DHT have a permissive role in the development of BPH but are not the sole cause. The androgen receptor remains at high levels in the prostate in aging men and specifically in BPH tissues, maintaining responsiveness to androgenic stimuli. The most common of the two 5α-reductase isoforms in the benignly enlarged prostate gland is type 2.
2. **d. Approximately 50% of cases of BPH in men who undergo surgery when younger than the age of 60 years are estimated to be inheritable.** There is significant evidence to suggest that some cases of BPH are familial, with autosomal dominant being the most likely inheritance pattern. An increased risk for BPH surgery exists mostly for men who come to BPH surgery when younger than the age of 60 years, and men with familial cases of BPH have larger glands than men with sporadic cases.
3. **b. Existing cases per 100,000 population at a distinct target date.** When studying diseases by descriptive or analytical epidemiologic methods, it is important to have a good understanding of the definitions that apply. Most epidemiologic terms are expressed as rates, which are the number of cases for persons expressed over the population. The definitions that are of relevance are incidence rates, which are equal to the number of people/100,000 population/year getting a certain disease; prevalence, which is the number of existing cases of the disease of interest/100,000 at a distinct target date; mortality rate, which is the number of deaths/100,000 population/year; and fatality, which equals the number of deaths due to the disease/number of diseased people.

4. **c. It is very uncommon in men younger than 30 years.** The autopsy prevalence of BPH has been studied as early as 1984 by Berry and colleagues. [a] Since then, many studies have been done on virtually all continents in many ethnic groups. It is astonishing that these studies find a very significant agreement in terms of the actual prevalence of histologic BPH or stromoglandular hyperplasia around the world. Stromoglandular hyperplasia or BPH is very uncommon in men younger than the age of 30 years, but then increases steadily in an almost linear manner. In fact, approximately 90% of men in their 80s have evidence of stromoglandular hyperplasia.
5. **d. The IPSS score has been translated and validated in many languages.** The IPSS symptom score is a seven-question, self-administered questionnaire that yields a total score ranging from 0 to 35 points. Men who score 0 to 7 points are classified as mildly symptomatic, those scoring from 8 to 19 points as moderately symptomatic, and those scoring from 20 to 35 points as severely symptomatic. The IPSS score addresses both voiding and storage, but not incontinence symptomatology. It is widely accepted that quantitative symptom scores are more important than, for example, urinary flow rate recordings. The IPSS score, the most widely utilized instrument, has been translated and culturally validated in many languages. Providers should abstain from filling in the questionnaire for their patients as it is validated as a self-reported symptom severity instrument.
6. **a. International studies show significant similarity in prostate volume in white, age-stratified men.** Prostate volume can relatively easily be assessed by transrectal ultrasonography (TRUS). TRUS has been found to be a reliable measure that is reproducible across examiners, in contrast to DRE, which is only poorly reproducible. MRI is very expensive and it yields in general a larger volume compared with TRUS measurements. Of note is the fact that international studies show significant similarity in regard to total and transitional zone prostate volume in white, age-stratified men.

7. **c. The intake of ethanol can decrease serum testosterone levels by a variety of mechanisms.** It is known that alcohol intake may decrease plasma testosterone levels by reducing production of and increasing clearance of testosterone. However, despite this hypothetical reason for a lower incidence, an inverse relationship has been described. The age-adjusted multivariate relative risks for undergoing surgery for BPH in men drinking more than three or four glasses of alcohol per day is lower than in age-matched controls. Of course, this could be due to a bias against surgery in patients who are heavy drinkers and therefore in poor health. It is interesting to note, however, that in the majority of studies, namely, four of five, a lower prevalence of BPH is found in men with cirrhosis compared with those without cirrhosis.
8. **d. Antidepressants, antihistamines, and bronchodilators increase the symptom score by several points.** There is only one study that systematically assessed the effect of medications on urinary symptoms and flow rate. Cold medications containing α-sympathomimetics tend to exacerbate lower urinary tract symptoms by the expected effect on the smooth muscle of the bladder outlet. Data from the Olmsted County Study of Urinary Symptoms and Health Status Among Men show that daily use of antidepressants, antihistamines, or bronchodilators is associated with a 2- to 3-point increase in the symptom score. However, only the daily use of antidepressants is associated with a decrease in the age-adjusted urinary flow rate.
9. **a. A clinically useful correlation exists between prostate volume and serum prostate-specific antigen (PSA) level.** In general there is an absence of useful baseline correlations between subjective and objective parameters such as symptoms, frequency, quality of life, and urinary flow rate measures of obstruction and prostate volume. However, symptom, bother, and interference with quality of life show excellent correlation with each other, and a clinically useful correlation exists between total and transition zone prostate volume and serum PSA in men with BPH.
10. **b. A longitudinal population-based study has the fewest biases and is the most useful type of study.** There are several ways to

study the natural history of BPH. One can look at watchful waiting cohorts or placebo-controlled groups of medication trials, or study population-based groups of men longitudinally over time. The latter is clearly the best way of studying the natural history of the disease because it incurs the fewest biases. However, it is also the most tedious and most expensive method. Placebo groups in medication trials clearly suffer from enrollment biases but do provide useful information.

11. **c. The higher the baseline score, the more of a drop is required for patients to subjectively feel improved.** The placebo response is partially a regression to the mean and partially an effect induced by the interaction between patient and doctor. The response is clearly dependent on the baseline severity score, with patients' higher scores having a larger decrease from baseline. The perception of subjective improvement has been shown to be dependent on the drop from baseline as well as on the baseline itself. For example, the higher the baseline score, the more of a drop from baseline is required for patients to have a subjective perception of improvement. Overall, a 3-point decrease is associated with a subjective perception of improvement.
12. **a. Depending on the population studied, incidence rates less than 5 to more than 130 cases/1000 man-years have been reported.** AUR has been studied during the past few years in population-based studies as well as in placebo-control groups from long-term treatment trials. The incidence rates differ significantly between different studies because of the inclusion and exclusion criteria and selection biases. Fortunately, AUR is a very clearly defined outcome and, thus, incidence rates can easily be calculated and compared.
13. **d. Age has been found to be the most significant risk factor for AUR in population-based studies.** In population-based studies such as the Olmsted County Study of Urinary Symptoms and Health Status Among Men, age is the most significant predictor of AUR. Data from placebo groups of long-term medical treatment trials demonstrate that serum PSA is the most powerful predictor of AUR together with prostate volume. Although this appears on the

surface to be a contradiction, it can be relatively easily explained by the fact that in BPH treatment trials, elderly men with already an existing diagnosis of BPH are enrolled. Thus, age plays a lesser factor in terms of predicting AUR. In population-based studies in which men stratified by age are followed during long periods of time, age plays a more significant factor compared with PSA.

14. **b. AUR is a harder and more objective end point compared with surgery.** Incidence rates of surgery vary significantly across geographic regions and patients with different ethnic backgrounds. Depending on the interaction between patient and physician, the physician can convince the patient to undergo surgery or, based on the patient's comorbidities, talk him out of surgery. The same cannot be said for urinary retention. It is clear that a vast majority of patients do not require surgery in the course of their disease but, rather, can be treated effectively with reassurance alone or medication.

Chapter Review

1. BPH is characterized by an increased number of epithelial and stromal cells, not an increase in their size.
2. Androgens are required for normal cell proliferation and differentiation and actively inhibit cell death.
3. Serum estrogen levels increase in men with age.
4. Early periurethral nodules are stromal; transition zone proliferation is glandular.
5. Prostatic stroma represents 40% of the gland. Smooth muscle is a prominent component of the stroma.
6. Autonomic system overactivity may contribute to lower urinary tract symptoms in men with BPH; the alpha 1a receptor is the most abundant form in the prostate.
7. Symptoms that use the AUA Symptom Index are classified as mild if the score is 0 to 7, moderate if it is 8 to 19, and severe if it is 20 to 35. A change of 3 points or more from time to time is subjectively discernible.

8. Men and women experience a decrease in maximum urinary flow rate as they age.
9. Bladder fibrosis is seen in both sexes with advancing age.
10. After spontaneous acute urinary retention (AUR), 15% of patients will have another episode and three fourths will undergo surgery; after precipitated AUR, 9% will have another episode and 26% will undergo surgery.
11. A significant portion of male lower urinary tract symptoms is related to age-related detrusor dysfunction and other conditions unrelated to the prostate.
12. DHT, the most potent androgen in the prostate, and androgen receptors remain high with age.
13. Androgen withdrawal results in apoptosis of prostate cells.
14. Estrogen receptors are found in the prostate and may play a role in BPH.
15. The size of the prostate does not correlate with the degree of obstruction.
16. Trabeculation is due to an increase in detrusor collagen.
17. A maximum flow rate less than 10 mL/sec in the male indicates a high probability of obstruction.
18. There is no relationship between vasectomy and BPH; however, there is a positive relationship between lack of physical activity, obesity, BMI, and LUTS/BPH.
19. Hydronephrosis is found in 7.6% of patients having surgery for BPH, one third of whom have renal insufficiency.
20. Some cases of BPH are familial, with autosomal dominant being the most likely inheritance pattern. Patients with familial BPH tend to have larger glands than those with sporadic BPH.
21. Cold medications containing α-sympathomimetics tend to exacerbate lower urinary tract symptoms by the expected effect on the smooth muscle of the bladder outlet.
22. A clinically useful correlation exists between total and transition zone prostate volume and serum PSA in men with BPH.

[a] Sources referenced can be found in *Campbell-Walsh-Wein Urology, 12th Edition,* on the Expert Consult website.

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Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

Paolo Capogrosso, Andrea Salonia, and Francesco Montorsi

Keywords

benign prostatic hyperplasia; lower urinary tract symptoms; medical therapy

Questions

1. Lower urinary tract symptoms (LUTSs) may arise as a consequence of:
    a. benign prostatic enlargement.
    b. bladder outlet obstruction.
    c. neurologic disease.
    d. detrusor underactivity.
    e. all of the above.
2. Medications that may exacerbate LUTSs include:
    a. loop diuretics.
    b. α adrenergic antagonists.
    c. β-3 adrenergic agonists.
    d. muscarinic receptors agonists.
    e. phytotherapy.

3. Digital rectal examination at the time of the assessment of patients with LUTSs:
    a. should always be performed with the only aim to rule out prostate cancer.
    b. allows estimation of prostate volume, which is useful for planning further management.
    c. does not provide reliable information, and therefore should not be performed.
    d. may be used to obtain prostatic secretion.
    e. has the main goal of estimating the rectal tone.
4. Frequency-volume charts:
    a. should always be used when patients with LUTSs are being assessed.
    b. should cover at least a 72-hour period.
    c. are particularly useful to assess obstructive symptoms.
    d. allows estimation of the average postvoid residual volume.
    e. should be requested only when a diagnosis of detrusor overactivity has been established.
5. It is advisable in a man with LUTSs/benign prostatic hyperplasia and a slightly elevated creatinine level to perform:
    a. transurethral resection of the prostate.
    b. an intravenous pyelogram.
    c. renal ultrasound imaging.
    d. a urodynamic study.
    e. flexible cystoscopy.
6. Which test should always be considered in a man with prevalent storage symptoms and a history of smoking?
    a. Urinalysis.
    b. Serum prostate-specific antigen (PSA) test.
    c. Urine cytology.
    d. Frequency-volume charts.
    e. Noninvasive urodynamic tests.
7. Serum PSA testing should be requested in which of the following clinical scenarios?
    a. A patient older than 75 years with a family history of prostate cancer.

b. A 55-year-old patient with negative digital rectal examination findings and no previous PSA measurements.
c. Serum PSA should be performed every case to rule out prostate cancer.
d. A patient with severe symptoms regardless of age.
e. A patient with a recent episode of macrohematuria.

8. What is the gold standard for the assessment of bladder outlet obstruction?
a. Invasive urodynamic tests.
b. Uroflowmetry.
c. Transrectal prostate ultrasound imaging.
d. Flexible urethrocystoscopy.
e. Cystourethrogram.

9. What is the suggested first treatment approach for a 50-year-old patient with an International Prostate Symptom Score (IPSS) score of 7 without risk factors for disease progression?
a. Phytotherapy.
b. α Adrenergic blockers.
c. Watchful waiting.
d. No treatment.
e. β3 Adrenergic agonists.

10. Which of the following strategies should be part of watchful waiting management?
a. Regular patient monitoring.
b. Education and reassurance.
c. Fluid management.
d. Concurrent medication adjustment.
e. All of the above.

11. Which of the following is an indication for surgical treatment in patients with LUTSs?
a. Severe symptoms.
b. Postvoid residual volume urine of 300 mL or more.
c. Recurrent episodes of acute urinary retention.
d. Elevated PSA level.
e. Lack of response to an α-blocker.

12. Detrusor overactivity:

a. is always an indicator of overactive bladder syndrome.
b. is rarely observed in patients with benign prostatic enlargement.
c. may be observed in up to 83% of men with severe bladder outlet obstruction.
d. should always be reduced by LUTS treatment.
e. None of the above sentences are true.

13. Which of the following receptors is not relevant for LUTS treatment?
    a. $\alpha\text{-}1_a$.
    b. $\alpha\text{-}1_b$.
    c. $\alpha\text{-}1_d$.
    d. β-3.
    e. β-1.
14. Which of the following is considered a uroselective α adrenergic blocker?
    a. Terazosin.
    b. Sildenafil.
    c. Tamsulosin.
    d. Doxazosin.
    e. None of them.
15. In the first placebo-controlled phase III trial, terazosin showed a symptom-reduction rate of approximately:
    a. 22%.
    b. 13%.
    c. 85%.
    d. 40%.
    e. 8%.
16. One of the most commonly reported side effects with terazosin therapy is:
    a. retrograde ejaculation.
    b. anejaculation.
    c. headache.
    d. dyspepsia.
    e. dizziness.

17. Pooled data analysis of randomized trials showed that the relative risk of postural hypotension with doxazosin treatment as compared with placebo is:
    a. approximately 2.7.
    b. approximately 5.6.
    c. approximately 7.8.
    d. approximately 1.1.
    e. not different.
18. The Medical Therapy of Prostatic Symptoms (MTOPS) trial showed that treatment with doxazosin was associated with:
    a. reduced risk of clinical progression as compared with placebo.
    b. increased risk of acute urinary retention as compared with placebo.
    c. no change in the risk of clinical progression as compared with placebo.
    d. reduced risk of surgical treatment as compared with finasteride.
    e. none of the above.
19. Which of the following drugs needs dose titration at first prescription:
    a. Finasteride.
    b. Alfuzosin.
    c. Terazosin.
    d. Doxazosin gastrointestinal therapeutic system.
    e. Tamsulosin.
20. An effect of tamsulosin on patient symptoms should be expected after approximately:
    a. 1 month of treatment.
    b. 6 months of treatment.
    c. 1 week of treatment.
    d. 3 months of treatment.
    e. 1 day of treatment.
21. What is the magnitude of the change of the bladder outlet obstruction index (BOOI) after treatment with silodosin?
    a. −30.4.
    b. −5.4.

c. −11.1.
d. −2.1.
e. Treatment with α adrenergic blockers is never associated with an improvement of urodynamic parameters.

22. Naftopidil has demonstrated high affinity for which of the following receptors?
    a. α-1_b.
    b. α-1_a.
    c. α-1_a and α-1_b.
    d. β3.
    e. α-1_d.
23. Randomized active-controlled trials comparing the effect of different α adrenergic blockers demonstrated that:
    a. tamsulosin at a dose of 0.4 mg is as effective as silodosin at a dose of 8 mg in terms of symptom relief.
    b. alfuzosin at a daily dose of 10 mg has lower overall efficacy as compared with tamsulosin at a dose of 0.4 mg.
    c. tamsulosin at a dose of 0.4 mg showed the greatest BOOI reduction among all available α adrenergic blockers.
    d. uroselective α adrenergic blockers are generally more effective than doxazosin and terazosin.
    e. there is no clinically relevant difference in the efficacy and tolerability profile among α adrenergic blockers.
24. Physicians should warn patients regarding the risk of first-dose hypotension when prescribing:
    a. silodosin.
    b. tamsulosin.
    c. terazosin.
    d. alfuzosin.
    e. all of the above.
25. What is the likely mechanism for dizziness after α-1 adrenergic blocker therapy?
    a. Reduced cardiac output.
    b. Central nervous system.
    c. Carotid baroreceptor.

d. Both a and c.
e. None of the above.

26. Before they undergo cataract surgery, patients should be advised to:
    a. discontinue treatment with α-blockers to reduce the risk of intraoperative floppy iris syndrome (IFIS).
    b. report the use of α-blockers to the ophthalmologist.
    c. discontinue treatment with tamsulosin and switch to a nonuroselective α-blocker.
    d. reduce the dosage of the α-blocker at least 1 month before surgery.
    e. do none of the above.
27. Which of the following effects should be expected after 1 month of treatment with finasteride?
    a. Significant decrease in serum dihydrotestosterone (DHT) level.
    b. Significant decrease in prostate volume.
    c. Significant increase in the peak urinary flow.
    d. 70% decrease in serum PSA level.
    e. None of the above.
28. What is the expected decrease in prostate volume with 5-α reductase inhibitors (5-ARIs)?
    a. 15% to 30%.
    b. More than 50%.
    c. 10%.
    d. 80%.
    e. 5%.
29. Regarding the safety profile of 5-ARIs:
    a. 5-ARIs may increase the risk of diabetes mellitus.
    b. patients treated with 5-ARIs showed higher risk of cardiovascular events.
    c. 5-ARIs may increase the overall risk of prostate cancer.
    d. patients should be warned regarding a higher risk of depressive symptoms associated with 5-ARI treatment.
    e. none of the above are true.
30. Tolterodine should not be prescribed to a patient with:
    a. a total IPSS score suggestive of severe symptoms.

b. postvoid residual volume greater than 200 mL.
c. evidence of detrusor overactivity.
d. low cystometric capacity.
e. prostate volume greater than 50 mL.

31. Who is the best candidate for treatment with mirabegron?
    a. A 75-year-old patient with prevalent storage symptoms.
    b. A 65-year-old patient with moderate LUTSs and a postvoid residual volume of 250 mL.
    c. A 70-year-old patient with moderate LUTSs and a previous episode of acute urinary retention.
    d. A 45-year-old patient with mild LUTSs and erectile dysfunction.
    e. None of the above.
32. Which of the following is considered a possible biologic effect of phosphodiesterase type 5 inhibitors (PDE5is) on the lower urinary tract?
    a. Relaxation of smooth muscle at the level of the prostate and bladder neck.
    b. Decreased blood flow at the level of the bladder and prostate resulting in decreased local inflammation.
    c. Increased activity of afferent nerves in the bladder wall.
    d. Inhibition of the 5α-reductase type 2 isoenzyme at the level of the prostate.
    e. All of the above.
33. What is the effect of PDE5i monotherapy in patients with LUTSs?
    a. There is no evidence of a significant reduction in the IPSS score with PDE5i monotherapy alone.
    b. PDE5i monotherapy resulted in a significant increase in the peak urinary flow after 1 month of treatment.
    c. PDE5i monotherapy has been shown to significantly decrease the IPSS score as compared with placebo.
    d. Tadalafil is the only PDE5i that has resulted in relief of symptoms in randomized trials.
    e. PDE5i monotherapy can decrease the risk of acute urinary retention in the long term.

34. Pooled data analysis has shown that the mean difference in the posttreatment IPSS between PDE5is and placebo is:
    a. approximately 2 to 4 points.
    b. approximately 6 to 8 points.
    c. approximately 1 to 2 points.
    d. not significant.
    e. none of the above.
35. The MTOPS trial showed that:
    a. the combination of doxazosin and finasteride is superior to placebo in reducing the risk of clinical progression.
    b. doxazosin is not superior to placebo in reducing the risk of clinical progression.
    c. finasteride alone does not reduce the risk of subsequent invasive therapy as compared with placebo.
    d. combination therapy is not superior to doxazosin monotherapy in terms of symptom relief.
    e. doxazosin is superior to placebo in reducing the risk of subsequent invasive therapy.
36. The Combination of Avodart and Tamsulosin (CombAT) study showed that:
    a. tamsulosin monotherapy significantly reduces the risk of acute urinary retention.
    b. dutasteride alone reduces the risk of surgical treatment as compared with tamsulosin.
    c. combination therapy is not superior than dutasteride monotherapy in reducing the risk of disease progression.
    d. tamsulosin is superior to dutasteride in reducing the IPSS score at long-term assessment.
    e. None of the above.
37. According to international clinical guidelines, a patient with a large prostate volume and predominantly bothersome voiding symptoms should be counseled for:
    a. surgical treatment.
    b. α-blocker plus 5-ARI therapy.
    c. α-blocker monotherapy.
    d. 5-ARI monotherapy.

e. α-blocker plus antimuscarinic drug therapy.

38. The definitive mechanism of action for *Serenoa repens* is:
    a. inhibition of 5α-reductase.
    b. inhibition of cyclooxygenase.
    c. inhibition of lipoxygenase.
    d. prostate smooth muscle relaxation.
    e. inconclusive.
39. The results of placebo-controlled trials assessing treatment with *S. repens* showed:
    a. a significant reduction of symptom scores.
    b. a reduction of the IPSS score only at triple the usual dose.
    c. no difference in the urinary flow parameters but significant relief symptoms.
    d. no difference in both symptoms and urinary flow parameters as compared with placebo.
    e. None of the above.
40. In a patient with a first episode of acute urinary retention, what is the probability of a successful trial without catheter?
    a. Approximately 60%.
    b. Less than 50%.
    c. Approximately 10%.
    d. Very low without concomitant α-blocker therapy.
    e. More than 80%.

Answers

1. **e. All of the above.** LUTSs are considered a clinical manifestation with a multifactorial pathophysiology arising from different conditions, including benign prostatic enlargement, bladder outlet obstruction, neurologic disease, and bladder dysfunction.
2. **a. Loop diuretics.** Loop diuretics may increase urinary output, exacerbating symptoms in patients with bladder outlet obstruction.
3. **b. Allows estimation of prostate volume, which is useful for planning further management.** Digital rectal examination should always be performed at the first assessment of patients with LUTS.

It can provide an estimate of prostate volume and at the same time rule out locally advanced prostate cancer.

4. **b. Should cover at least a 72-hour period.** A 3-day period is considered a reasonable compromise to increase the accuracy of frequency-volume charts without decreasing patient adherence.
5. **c. Renal ultrasound imaging.** International clinical guidelines suggest upper urinary tract imaging in the case of elevated serum creatinine level.
6. **c. Urine cytology.** This test should be considered in patients presenting with storage symptoms and a history of smoking to rule out urothelial neoplasms.
7. **b. A 55-year-old patient with negative digital rectal examination findings and no previous PSA measurements.** Among patients with LUTSs, PSA testing should be suggested if a diagnosis of prostate cancer would change the clinical management.
8. **a. Invasive urodynamic tests.** The diagnosis of bladder outlet obstruction relies on urodynamic tests.
9. **c. Watchful waiting.** Patients with mild symptoms without factors suggesting a high risk of disease progression (e.g., large postvoid volume, large prostate volume) should be counseled for conservative management.
10. **e. All of the above.** Watchful waiting is mainly based on lifestyle modifications, education, and reassurance and monitoring.
11. **c. Recurrent episodes of acute urinary retention.** Surgical treatment of LUTSs is indicated for patients with recurrent acute urinary retention because they may not benefit from medical treatment.
12. **c. May be observed in up to 83% of men with severe bladder outlet obstruction.** Detrusor overactivity is extremely common in the case of bladder outlet obstruction; however, LUTS treatment may not reduce detrusor overactivity.
13. **e. β-1.** Among adrenergic receptors, the β-1 subfamily is poorly represented in the lower urinary tract and is not targeted by any of the currently available treatments for LUTSs.

14. **c. Tamsulosin.** Uroselective agents have high affinity for the α-1_a and α-1_d receptors; as such, they have greater selectivity for prostatic and urethral tissue.
15. **a. 22%.** The first phase II multicenter, placebo-controlled trial showed a 22.4% reduction in symptom score from the baseline for patients treated with terazosin.
16. **e. Dizziness.** As a consequence of the vascular side effects associated with the inhibition of α-1_b receptors, dizziness is frequently reported after treatment with terazosin and doxazosin.
17. **a. Approximately 2.7.** A meta-analysis of 10 placebo-controlled trials showed that the relative risk of postural hypotension associated with doxazosin treatment was 2.72 (95% confidence interval 1.21–6.15).
18. **a. Reduced risk of clinical progression as compared with placebo.** In the MTOPS trial, doxazosin therapy resulted in a significant decrease in the risk of clinical progression as compared with placebo, although it was inferior to both finasteride monotherapy and combination therapy.
19. **c. Terazosin.** Given the high risk of first-dose hypotension, nonuroselective α-blockers need dose titration over the first few weeks of treatment.
20. **c. 1 week of treatment.** Randomized trials have shown that tamsulosin leads to symptom relief after 1 week of treatment and to changes in urinary flow parameters after only a few hours.
21. **a. –30.4.** In a recent study pooling data from trials assessing urodynamic changes associated with α-blocker treatment, silodosin showed the greatest BOOI reduction as compared with the other compounds.
22. **e. α-1_d.** Naftopidil is the only developed compound with a distinct selectivity for the α-1_d receptor subtype.
23. **a. Tamsulosin at a dose of 0.4 mg is as effective as silodosin at a dose of 8 mg in terms of symptom relief.** Data from two meta-analyses of randomized controlled trials comparing silodosin with tamsulosin showed no difference in terms of efficacy.

24. **e. All of the above.** Despite the lower risk of vascular side effects associated with uroselective α-blockers, patients should be always warned regarding the risk of first-dose hypotension.
25. **b. Central nervous system.** The α-1–mediated dizziness and asthenia are likely due to effects at the level of the central nervous system.
26. **b. Report the use of α-blockers to the ophthalmologist.** IFIS cannot be prevented, but the ophthalmologist should be notified regarding the use of α-blockers so as to properly manage complications.
27. **e. None of the above.** As a type 1 5-ARI, finasteride does not lower the serum level of DHT. Moreover, 1 month of treatment is not enough to observe a significant change in prostate volume and urinary flow parameters. Conversely, PSA level may change after 1 month of treatment.
28. **a. 15% to 30%.** Randomized trials assessing treatment with dutasteride and finasteride have shown a prostate volume reduction ranging from 15% to 30% after 6 to 24 months of treatment.
29. **d. Patients should be warned regarding a higher risk of depressive symptoms associated with 5-ARI treatment.** Large prospective trials have shown a higher prevalence of depressive symptoms among patients treated with 5-ARIs.
30. **b. Postvoid residual volume greater than 200 mL.** All trials investigating antimuscarinic drugs included only patients with a postvoid residual volume lower than 200 mL.
31. **a. A 75-year-old patient with prevalent storage symptoms.** Given its more favorable tolerability profile as compared with antimuscarinic drugs, mirabegron therapy could be preferred in old men with prevalent storage symptoms.
32. **a. Relaxation of smooth muscle at the level of the prostate and bladder neck.** This effect is associated with the increase in nitric oxide signaling from nerve fibers close to the muscle fibers.
33. **c. PDE5i monotherapy has been shown to significantly decrease the IPSS score as compared with placebo.** Several randomized trials demonstrated significant relief of symptoms associated with

PDE5i monotherapy. Conversely, there is no evidence of improved urinary flow parameters with PDE5is alone.

34. **a. Approximately 2 to 4 points.** Meta-analyses of placebo-controlled trials have shown a mean difference in the posttreatment IPSS score, between PDE5is and placebo, ranging from 2 to 4 points.
35. **a. The combination of doxazosin and finasteride is superior to placebo in reducing the risk of clinical progression.** The MTOPS trial demonstrated that combination therapy of an α-blocker and a 5-ARI can significantly reduce the risk of disease progression at long-term follow-up. This effect was greater than with both placebo and monotherapy.
36. **b. Dutasteride alone reduces the risk of surgical treatment as compared with tamsulosin.** The CombAT study showed a greater reduction in the risk of surgical treatment for patients treated with dutasteride as compared with tamsulosin.
37. **b. α-Blocker plus 5-ARI therapy.** Treatment algorithms suggest a combination of an α-blocker and a 5-ARI to treat patients with large gland volumes and bothersome voiding symptoms.
38. **e. Inconclusive.** Although experimental data have suggested numerous possible mechanisms of actions for the phytotherapeutic agents, it is uncertain which, if any, of these proposed mechanisms is responsible for the clinical responses.
39. **d. No difference in both symptoms and urinary flow parameters as compared with placebo.** Although meta-analysis of randomized trials of *S. repens* did not show significant relief of LUTSs, it has been recognized that those results could be affected by the heterogeneity of the extracts used in the different trials and may not be applicable to all marketed products.
40. **a. Approximately 60%.** Prospective studies have shown that a successful trial without catheter after a first episode of acute urinary retention could be achieved in 60% of cases.

Chapter Review

1. Assessing symptom severity is the first step in the workup of patients with LUTSs.
2. The PSA test should be considered for patients with a life expectancy longer than 10 years.
3. Ultrasound imaging and uroflowmetry should be performed for patients with moderate to severe symptoms.
4. Invasive urodynamic tests should be considered if bladder motility alterations (e.g., detrusor overactivity, detrusor underactivity) are suspected.
5. Conservative management is the preferred approach for patients with mild symptoms and without complicating factors.
6. Watchful waiting is based on patient education, lifestyle interventions, and disease monitoring.
7. Lifestyle modifications include proper daily fluid intake, tea, caffeine, and alcohol avoidance/restriction, and concurrent medication adjustment.
8. Patients should be followed up yearly to detect worsening of symptoms or the occurrence of complications.
9. α1-Blockers are the first line treatment for patients with moderate to severe LUTSs.
10. The uroselective α1-blockers tamsulosin, silodosin, and alfuzosin showed the best tolerability profile.
11. 5-ARIs are effective in reducing the risk of disease progression and the need for invasive treatment.
12. Patients with predominant storage symptoms may benefit from a combination therapy with antimuscarinic drugs.
13. PDE5is are effective in relieving LUTSs, either alone or in combination with α1-blockers.
14. There is no consensus on the efficacy of phytotherapy for LUTS treatment, although encouraging results have been published for the hexane extract of *S. repens* and for the combination of *S. repens* with lycopene and selenium.
15. Acute urinary retention (AUR) is a common complication of benign prostatic hyperplasia progression.
16. A trial without catheter (TWOC) is the preferred management after initial decompression by bladder catheterization.

17. The use of α1-blockers is recommended to increase the odds of a successful TWOC.
18. Dutasteride and finasteride have been shown to significantly decrease the risk of AUR in patients with moderate to severe LUTSs.

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Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

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Keywords

Benign prostatic hyperplasia; BPH; convective radiofrequency water vapor thermal therapy; lower urinary tract symptoms; LUTS; minimally invasive treatments; MIST; prostatic urethral lift; PUL; transurethral resection of the prostate; TURP

Questions

1. In the conduct of performing a transurethral resection of the prostate (TURP) on a 65-g prostate using a continuous flow instrument, the urologist notes a disturbing amount of dark blood from the rectoscope efflux. In carefully and repeatedly observing the prostate fossa he notes no obvious bleeding sites. The likely etiology of such blood is:
    a. venous bleeding from prostatic fossa.
    b. unstable blood pressure resulting in intermittent arterial bleeding.
    c. inadvertent use of chromogen dyes (methylene blue or indocarmine) by anesthesia.
    d. perforation of the bladder.

e. previous unrecognized coagulopathy.

2. A patient is 6 weeks post-TURP for long-standing obstruction and detrusor overactivity; voices marked concern over his persistent symptoms, wondering if there was an incomplete resection. A urine culture is negative. The next best sequence of care is:
    a. measure post void residual, provide reassurance, and monitor symptom improvement.
    b. measure post void residual volume and schedule formal urodynamics.
    c. restart a course of 5-alpha reductase inhibitors (5ARIs) and antibiotics.
    d. perform immediate cystoscopy looking for residual tissue.
    e. prescribe 90-day supply of phenazopyridine HCl
3. In the course of performing a monopolar TURP under spinal block, the nurse anesthetist reports the patient is suddenly markedly confused, brachycardic, and hypertensive. An emergent serum sodium level is 110 mEq/L. The following are the next best steps:
    a. Switch to bipolar TURP equipment and finish the case.
    b. Drop the height of the irrigating fluid to <60 cm above the patient, control bleeding, terminate the case, administer furosemide, and begin a calculated dose of hypertonic saline.
    c. Convert to an open simple prostatectomy
    d. Measure serum levels of illicit drugs and ammonia.
    e. Swap irrigating fluid form glycine to sorbitol.
4. Risk factors for development of the TUR syndrome include:
    a. prostate volume less than 45 g.
    b. failure to use isotonic, iso-osmolar irrigating solution and the bipolar electroresection system.
    c. underlying hepatic dysfunction.
    d. irrigating fluid above the patient should be ideal be >60 cm.
    e. preoperative use of opioids.
5. Transurethral resection (TUR) syndrome is caused by:
    a. absorption of fluid during procedures such as holmium laser enucleation of the prostate (HoLEP) and bipolar TURP.
    b. absorption of non–sodium-containing irrigating fluid, leading to an acute dilutional hyponatremia.

c. irrigating fluid placed at a less than ideal height above the patient.
d. a serum sodium of greater than 130 mEq/L.
e. intraoperative ureteral injury.

6. TURP should begin with resection of the:
   a. apical portion of the prostate.
   b. prostate floor.
   c. bladder neck.
   d. median lobe, if present.
   e. anterior portion of the prostate.
7. Which of the following is TRUE of bipolar compared with monopolar TURP?
   a. In a meta-analysis of patients undergoing bipolar TURP, authors concluded that by treating 50 patients with bipolar TURP, one case of TUR syndrome could be prevented.
   b. A relative risk of 0.53 for blood transfusion with bipolar resection was found in meta-analysis.
   c. Improved visualization during bipolar TURP may also lead to a decrease in capsular perforations and operating time.
   d. Late complications such as bladder neck contracture and need for retreatment of benign prostatic hyperplasia (BPH) do not appear to be much different from those found with conventional TURP.
   e. All of the above.
8. Sham studies on urologic procedures for lower urinary tract symptoms (LUTS) due to BPH:
   a. frequently show statistically significant decreases in AUASS (American Urological Association Symptom Score).
   b. are poorly tolerated by the patient.
   c. have significant side effects and should not be performed as part of research.
   d. have never shown a statistically significant improvement in objective measures such as peak urinary flow.
   e. have never been performed.
9. Which of the following is TRUE regarding transurethral incision of the prostate (TUIP)?

a. It commonly results in TUR syndrome.
b. It is generally only used in prostates larger than 60 mL.
c. It causes retrograde ejaculation in 80% of cases.
d. It results in removal of a large volume of prostate adenoma.
e. It may have a lower rate of ejaculatory dysfunction in patients when done unilaterally.

10. With regard to laser safety, which of the following statements is correct?
    a. Eye protection is required for the surgeon only.
    b. All windows or wall openings from the operating room (OR) must be covered.
    c. Signs denoting that a laser is in use need only be displayed on the most commonly used door for that OR.
    d. Eye protection is required only when a video camera is not used during the case.
    e. All laser energy is readily absorbed by air/irrigating fluid, making it safe to use in the OR.
11. Patients on anticoagulation who undergo photoselective vaporization of the prostate (PVP) have an increased risk of:
    a. erectile dysfunction.
    b. blood transfusion.
    c. TUR syndrome.
    d. ejaculatory dysfunction.
    e. time in the hospital after procedure.
12. When using the prostate lift for treatment of LUTS due to BPH, implants are placed where in the prostate anatomically?
    a. Anterolaterally
    b. Posterolaterally
    c. Anteriorly
    d. Posteriorly
    e. In the peripheral zone

Answers

1. **a. Venous bleeding from prostatic fossa.** Venous bleeding can be more vexing and generally appears darker in color than arterial bleeding. Venous bleeding will often disappear with a full bladder as it applies pressure to the prostatic fossa. Identifying venous sinuses may be difficult due to the lack of persistent bleeding with rapid irrigation influx and controlling of the bleeding with fulguration may be even more challenging. Once again, the cutting loop may be used to temporarily tamponade any bleeding with precise fulguration. If the venous bleeding is unable to be completely controlled (as is frequently the case) a catheter may be inserted, the balloon overinflated and pulled to traction for tamponade of venous bleeding.
2. **a. Measure post void residual, provide reassurance, and monitor symptom improvement.** Patients with long-standing obstruction (particularly those with urgency and frequency preoperatively) will often experience a continuation or exacerbation of these symptoms in the postoperative period. If proper bladder emptying can be verified, an anticholinergic or beta-3 agonist medication during this time may help the patient feel more comfortable. It is our practice to warn those men with preoperatively documented detrusor overactivity that patience will be required in the months following surgery to see if this will resolve. Such caution goes a long way in assuring patients to adopt a strong coping mechanism rather than a polypharmacy approach.
3. **b. Drop the height of the irrigating fluid to <60 cm above the patient, control bleeding, terminate the case, administer furosemide, and begin a calculated dose of hypertonic saline.** Patient is likely experiencing TUR syndrome; thus rapid termination is needed. The height of the irrigating fluid above the patient should be carefully chosen. Investigators have demonstrated that the ideal height of the fluid was 60 cm above the patient. This appears to be the minimal height to maintain good vision but also not lead to excessive systemic fluid absorption. Increasing the height 10 cm above this leads to increased pressure in the prostatic fossa and a greater than twofold increase in systemic fluid absorption. Diagnosis of this condition is made by assessment of

neurologic status and comparison to laboratory values. Serum sodium should be obtained in long, large resections postoperatively (or intraoperatively if concern exists). A serum sodium of less than 120 mEq/L indicates a significant dilution and may lead to coma or seizures. Transient visual disturbances or blindness indicate central nervous system (CNS) toxicity and are obviously very distressing to all the parties involved. If profound central nervous symptoms are noted, judicious administration of hypertonic saline should be instituted and formulas exist to help guide this resuscitation as overly rapid correction of hyponatremia may lead to a demyelinating lesion of the brain (central pontine myelinolysis).

4. **b. Failure to use isotonic, iso-osmolar irrigating solution and the bipolar electroresection system.** The absorption of the hypo-osmolar irrigating fluid leads to an acute dilutional hyponatremia with resulting neurologic changes (confusion, nausea, vomiting, visual changes, hypertension, tachypnea, and bradycardia). Now with the use of isotonic, iso-osmolar irrigating solution and the bipolar electroresection system this risk has theoretically been eliminated. While most authors agree that TUR syndrome is caused by dilutional hyponatremia, there have been alternate etiologies proposed. Excessive glycine absorption can lead to liberation of ammonia from metabolic pathways leading to immediate or delayed encephalopathic symptoms. Several steps can be taken to prevent this complication. Use of a bipolar resection method should certainly be considered. The height of the irrigating fluid above the patient should be carefully chosen. Use the minimal height to maintain good vision but also not lead to excessive systemic fluid absorption.
5. **b. Absorption of non–sodium-containing irrigating fluid, leading to an acute dilutional hyponatremia.** Absorption of non–sodium-containing irrigating fluid into the prostatic venous system that is exposed during resection is the etiology of the disease. This risk appears to be unique to monopolar TURP; other BPH techniques (such as bipolar TURP, HoLEP, and laser vaporization) use isotonic/iso-osmolar irrigating fluid such as normal saline. **The ideal height of irrigating fluid was determined to be 60 cm**

above the patient, as this balanced the benefits of visualization with systemic absorption. Heights above this level will lead to an increased systemic absorption. In general, symptoms of TUR syndrome begin with a serum sodium of less than 120 mEq/L. Ureteral injury is not associated with TUR syndrome.

6. **d. Median lobe, if present. Although many different plans for resection exist, resection of the median lobe (when present) is generally accepted as the first step.**
7. **e. All of the above.** All of the findings in a through d have been demonstrated in studies. The use of a sodium-containing iso-osmolar irrigating fluid has essentially eliminated the risk of TUR syndrome in bipolar TURP. **The "cut and seal" action of the technology improves intraoperative hemostasis with better visualization, leading to less blood transfusion and quicker operating times.** Differences in many late complications such as bladder neck contracture and need for retreatment have not been demonstrated in comparison to monopolar technology.
8. **a. Frequently show statistically significant decreases in AUASS (American Urological Association Symptom Score).** Multiple sham studies have been performed as part of clinical trials on transurethral microwave thermotherapy and prostate lift. Significant improvements in both AUASS and peak flow have been shown. Side effects of treatment are infrequent, and the sham procedures are well tolerated.
9. **e. It may have a lower rate of ejaculatory dysfunction in patients when done unilaterally. The procedure is relatively short and does not cause TUR syndrome. The procedure is only appropriate for small prostate glands (generally less than 30 mL), and no prostate adenoma is removed. Retro grade ejaculation occurs in up to 37% of patients.** Although this is controversial, most authors believe that the risk of retrograde ejaculation is lower if done unilaterally as opposed to bilaterally.
10. **b. All windows or wall openings from the operating room (OR) must be covered. Eye protection is required for all classes of lasers used in urology currently.** Eye protection should be utilized by the patient and all personnel in the room even if a video camera

is used during the case. Signs should be placed on all entries to the OR. Any and all openings to the OR from which laser energy could escape should be covered to preclude injury to persons outside of the OR. Although holmium laser energy is absorbed in irrigating fluid, potassium-titanyl-phosphate/lithium-triborate (KTP/LBO) laser energy is not readily absorbed in either fluid. Both of these lasers can damage the eye when outside the body as neither are readily absorbed/dispersed in air.

11. **e. Time in the hospital after procedure.** Patients who undergo PVP while on anticoagulation appear to be more likely to require longer times in the hospital. They also appear to require more continuous bladder irrigation and a longer time with a urethral catheter. Blood transfusions do not appear to be more frequent. TUR syndrome does not occur with PVP, as normal saline is used. Erectile dysfunction and ejaculatory dysfunction in patients on anticoagulation during PVP are not well studied.
12. **a. Anterolaterally.** The implants in the prostate lift system are placed anterolaterally to avoid the neurovascular bundles (posterolateral) and the prostate veins (anterior). **The implants work by primarily compressing the transition zone of the prostate and leading to an increased opening of the urethral lumen.**

Chapter Review

1. The incorporation of medical management and new technologies has changed which treatment options are selected for the treatment of LUTS/BPH.
2. LASER based treatments are increasingly used although multiple factors have affected acceptance. Multiple factors have changed the usual patient that advances to surgical management of BPH in the preceding decades. Increasingly these are more challenging clinical scenarios.
3. Patients should generally be trialed on medical management before proceeding to surgical intervention. Patients failing medical

management may be considered for surgical therapy.

4. Perioperative antibiotic coverage should not be neglected as serious adverse outcomes still occur.
5. Both patient and physician factors should be considered when selecting the appropriate surgical treatment for patients.
6. PUL is an option for patients with LUTS/BPH provided prostate volume <80 g when the clinician has verified the absence of a middle lobe.
7. Patients electing PUL or water vapor thermal ablation therapy should be warned that long-term durability is uncertain, though long-term uncontrolled data is available.
8. In the appropriate patients, PUL and water vapor thermal ablation therapy are less likely to impact erection and ejaculation than conventional therapies.
9. LASER treatments are the fastest growing option in treatment of LUTS/BPH but should be used as part of a “culture of safety” in the operating room.
10. HoLEP is a very effective treatment option with excellent results that are often comparable to those historically seen with open prostatectomy. There is a significant learning curve associated with the procedure and catastrophic complications (mostly due to the morcellator) have been observed.
11. Photoselective vaporization of the prostate is a growing and very safe treatment option for BPH/LUTS. Results in patients on anti-coagulation have been encouraging.
12. Thulium is the newest addition to the LASER family and has some theoretical advantages although scientific data is lacking.

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Simple Prostatectomy

Open and Robot-Assisted Laparoscopic Approaches

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Keywords

benign prostatic hyperplasia; open; robotic; simple prostatectomy

Questions

1. The major advantage of simple prostatectomy over transurethral resection of the prostate (TURP) in the management of prostatic adenoma includes:
    a. removal of the prostatic adenoma under direct vision.
    b. decreased risk of hypernatremia.
    c. shortened convalescence period.
    d. decreased perioperative hemorrhage.
    e. enhanced preservation of erectile function.
2. The suprapubic simple prostatectomy, in comparison to the retropubic simple prostatectomy, allows:
    a. direct visualization of the prostatic adenoma during enucleation.
    b. better visualization of the prostatic fossa after enucleation to obtain hemostasis.

c. easier management of a large median lobe and/or bladder calculi.
d. an extraperitoneal approach.
e. possible management of concomitant ureteral calculi.

3. The suprapubic approach to the prostatectomy is ideal for the patient with a large prostatic adenoma and:
   a. multiple small bladder calculi.
   b. total prostate-stimulating antigen (PSA) greater than 10.0 ng/mL.
   c. erectile dysfunction.
   d. symptomatic bladder diverticulum.
   e. presence of dilated renal pelvis.
4. The most appropriate definitive treatment options for the patient with a 120-g prostatic adenoma and a symptomatic bladder diverticulum are:
   a. radical retropubic prostatectomy with the fulguration of bladder diverticulum.
   b. long-acting α-adrenergic antagonist and prophylactic antibiotics.
   c. TURP followed by bladder diverticulectomy in 3 months.
   d. TURP and partial cystectomy.
   e. simple prostatectomy with bladder diverticulectomy.
5. The contraindications to simple prostatectomy include:
   a. multiple cores of Gleason 8 prostate cancer.
   b. bladder diverticulum.
   c. large bladder calculi secondary to obstruction.
   d. recurrent urinary tract infection.
   e. acute urinary retention.
6. Both retropubic and suprapubic simple prostatectomies:
   a. are performed in the space of Retzius.
   b. are ideal for patients with a large, obstructive prostatic adenoma and a concomitant, small bladder tumor.
   c. require no blood transfusion.
   d. cause no trauma to the urinary bladder.
   e. require control of the dorsal vein complex before the enucleation of an obstructive prostatic adenoma.

7. Compared with open simple prostatectomy, robot-assisted laparoscopic simple prostatectomy has:
    a. quicker operative time.
    b. longer hospital stay.
    c. decreased need for blood transfusion.
    d. shorter learning curve.
    e. decreased need for anesthesia.

Answers

1. **a. Removal of the prostatic adenoma under direct vision.** When compared with TURP, simple prostatectomy offers the advantages of lower retreatment rate and more complete removal of the prostatic adenoma under direct vision and avoids the risk of dilutional hyponatremia (the TUR syndrome).
2. **c. Easier management of a large median lobe and/or bladder calculi.** The major advantage of suprapubic simple prostatectomy over the retropubic approach is that it allows direct visualization of the bladder neck and bladder mucosa. As a result, this operation is ideally suited for patients with a large median lobe protruding into the bladder or a clinically significant bladder diverticulum.
3. **d. Symptomatic bladder diverticulum.** Suprapubic simple prostatectomy is ideally suited for patients with a large median lobe protruding into the bladder, a clinically significant bladder diverticulum that requires repair, or large bladder calculi. It also may be preferable for obese men, in whom it is difficult to gain direct access to the prostatic pseudocapsule and dorsal vein complex.
4. **e. Simple prostatectomy with bladder diverticulectomy.** Combined suprapubic simple prostatectomy and bladder diverticulum is the best option for the patient with a massive benign prostatic hyperplasia (BPH) and a symptomatic bladder diverticulum because of easy access to both prostatic adenoma and bladder diverticulum. If the prostatectomy is performed without the

diverticulectomy, incomplete emptying of the bladder diverticulum and subsequent, persistent infection may occur.

5. **a. Multiple cores of Gleason 8 prostate cancer.** Contraindications to simple prostatectomy include a small fibrous gland, the presence of significant prostate cancer, and previous prostatectomy or pelvic surgery that may obliterate access to the prostate gland.
6. **a. Are performed in the space of Retzius.** Both retropubic and suprapubic simple prostatectomies are performed in the space of Retzius. The dorsal vein complex does not have to be controlled before the enucleation of an obstructive prostatic adenoma during suprapubic simple prostatectomy.
7. **c. Decreased need for blood transfusion.** Compared with open simple prostatectomy, robot-assisted laparoscopic simple prostatectomy can be performed with smaller incisions with shorter hospital stay and decreased risk for perioperative hemorrhage and blood transfusion.

Chapter Review

1. Of the indications for simple prostatectomy—(a) acute urinary retention, (b) recurrent or persistent urinary tract infections, (c) significant symptoms of bladder outlet obstruction not responsive to medical therapy, (d) persistent gross hematuria from the prostate, (e) pathologic changes of the kidneys secondary to prostatic obstruction, and (f) bladder calculi—the only absolute indication for prostatectomy is pathologic changes of the kidneys secondary to prostatic obstruction. All the others are relative indications because they may on occasion be corrected without the need for simple prostatectomy.
2. One should not consider doing a simple prostatectomy for glands of less than 75 g. A simple prostatectomy on small glands carries a high complication rate.
3. Simple prostatectomy should be considered in patients who cannot be placed in the lithotomy position and who have a sufficiently large gland that can be enucleated. Small fibrous glands, the presence of

significant prostate cancer, previous prostatectomy, and pelvic surgery are contraindications to open simple prostatectomy.

4. Although a cystoscopic examination is not indicated for routine evaluation of obstructive voiding symptoms, one must estimate the size of the prostate adenoma preoperatively to schedule the patient appropriately. Cystoscopy may be a crucial component of that estimation. Moreover, in patients who have hematuria or a urethral stricture, or in whom one needs to evaluate a known bladder calculus or diverticulum, cystoscopy is indicated.
5. The risks of simple prostatectomy include urinary incontinence, erectile dysfunction, retrograde ejaculation, urinary tract infections, bladder neck contracture, urethral stricture, and the need for a blood transfusion.
6. Suprapubic simple prostatectomy is ideally suited for patients with a large median lobe protruding into the bladder, a clinically significant bladder diverticulum that requires repair, or large bladder calculi. It also may be preferable for obese men, in whom it is difficult to gain direct access to the prostatic pseudocapsule and dorsal vein complex.
7. Continuous bladder irrigation and blood transfusion are less frequently used following robot-assisted laparoscopic simple prostatectomy.